Preserving Exact Long-Range Recall in Compressed Transformers

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`github.com/kdrana/OLSAttentionExploration`

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Abstract

Serving large language models across long context horizons is critically bottlenecked by the Key-Value (KV) cache, whose linear footprint in sequence length T exhausts accelerator high-bandwidth memory (HBM) and restricts multi-tenant throughput. Post-training compression strategies have split into two competing paradigms: compiling historical context into fixed-size recurrent matrix states, or selectively evicting tokens from the KV cache. In this work, we demonstrate the fundamental geometric failure modes of both approaches and introduce a principled mechanism that resolves them without fine-tuning. First, we prove the **Rank-r Spectral Dilution Horizon**: any fixed-size recurrent state acts as a low-pass spectral filter whose energy allocation to an individual token decays as $\mathcal{O}(1/T)$, causing exact long-range recall to collapse to 0.0% beyond the local window despite capturing over 95% of attention output variance ($R^2 = 0.9505$) and preserving 100% general reasoning accuracy. Second, we uncover the **Query-Answer Asymmetry Principle**: existing selective token eviction schemes fail on complex entities because causal prefix scoring is blind to terminal prompt queries, while greedy top-k selection retains query-matching cue words while severing the adjacent multi-token payload. In response, we introduce **HALO** (**H**ybrid **A**ttention with **L**andmark **O**bservation-diffusion), which pairs an exact local sliding window with phase-rectified observation alignment, bottom-up geometric surprise, and an **Exponential Salience Halo** ($\kappa(\delta) = \exp(-0.10|\delta|), K = 16$) that diffuses importance contiguously across complete factual clauses, alongside a unified **HALO + BLUR** synthesis combining exact landmarks with compressed residual background statistics. Evaluated across three distinct foundation model families from independent organizations—Alibaba (`Qwen3.5-0.8B`), Meta (`Llama-3.2-1B`), and HuggingFace (`SmolLM2-1.7B`)—and across the canonical RULER 4-axis benchmark suite, HALO achieves 100.0% single-needle, 90.0% adversarial distractor, and 100.0% multi-hop variable tracking recall while compressing KV-cache memory by 16× to 25.6× with sub-60ms selection overhead.

1 Introduction

The rapid expansion of context windows in modern large language models has shifted the primary deployment bottleneck from compute throughput to GPU High-Bandwidth Memory (HBM) capacity and memory bandwidth.

1.1 The Linear Memory Bottleneck and Surgical Adaptation

In autoregressive Transformer inference [Vaswani et al., 2017], generating each new token requires attending across all previous key-value pairs. Consequently, the Key-Value cache (KV-cache) grows strictly linearly with context length T:

$$\text{Memory}_{\text{KV}}(T) = 2 \times L \times H_{\text{KV}} \times T \times d \times \text{bytes_per_element} \quad (1)$$

In a modern hybrid architecture like `Qwen/Qwen3.5-0.8B-Base` [Yang et al., 2024], its 6 full-attention layers require 384.0 MiB per sequence at $T = 32,768$ and explode to 3.00 GiB at $T = 262,144$ (12.0 KiB per token). In a pure 28-layer attention model such as `Qwen/Qwen2.5-1.5B`, the cache consumes over 1.79 GiB per sequence at 32k. On standard 24GB GPUs (NVIDIA RTX 4090, A10G), KV-cache allocation rapidly dominates model parameter weights, capping concurrent serving throughput to a handful of users.

While clean-slate subquadratic architectures (e.g., Mamba-2, Dao and Gu 2024; Gated DeltaNet, Yang et al. 2025) achieve constant $\mathcal{O}(1)$ memory during pretraining, practitioners cannot afford to discard trillions of tokens of established foundation model checkpoints. Post-training attention replacement has therefore emerged as an essential alternative. However, gradient-based token distillation (e.g., *SUPRA*, Mercat et al. 2024; *MOHAWK*, Bick et al. 2024; *LoLCATs*, Zhang et al. 2025) requires extensive calibration datasets and induces severe out-of-domain vocabulary drift. This motivates a non-destructive, mathematically grounded investigation into alternative memory paradigms for compressed Transformers.

1.2 Taxonomy of Alternatives to the Full KV-Cache

We categorize post-training alternatives to the full $\mathcal{O}(T)$ KV-cache into four distinct paradigms:

1. **Fixed-Size Recurrent & Test-Time Regression States ($\mathcal{O}(1)$ Matrix Memory):** Replaces token histories with fixed-dimensional associative matrices $S_t \in \mathbb{R}^{d\times d}$ or $r\times r$, including Linear Attention [Katharopoulos et al., 2020], Mamba-2 [Dao and Gu, 2024], Gated DeltaNet [Yang et al., 2025], Test-Time Training [Sun et al., 2024], Titans [Behrouz et al., 2025], and our closed-form Ordinary Least Squares / Basis-Linear Unitary Recurrence (**BLUR**).
2. **Head- and Latent-Space KV Compression:** Compresses head dimensions or projects key-value states into low-rank latent spaces (DeepSeek-V2 MLA, DeepSeek-AI 2024). However, Rotary Position Embeddings (RoPE, Su et al. 2024) prevent direct low-rank compression without decoupling rotary components.
3. **Windowed Sinks and Online Causal Eviction:** Discards historical tokens using causal importance heuristics. *StreamingLLM* [Xiao et al., 2024] preserves initial attention sink tokens and a local window. H_2O [Zhang et al., 2023] accumulates prefix query-attention mass. However, because prefix tokens have not seen the generation prompt, causal accumulation is blind to the user's ultimate question.
4. **Observation-Window Selection & Contiguous Salience Diffusion:** Employs observation-window queries at the end of the prompt to identify salient historical clusters (*SnapKV*, Li et al. 2024; *Quest*, Tang et al. 2024). While effective for isolated keywords, standard top-k selection severs adjacent multi-token identifiers, a failure mode resolved by our **HALO** architecture and its unified **HALO + BLUR** variant.

1.3 Summary of Core Findings

1. **The Spectral Dilution Horizon of Matrix States:** Even when recurrent operators explain $> 95\%$ of attention variance ($R^2 = 0.9505$) via closed-form Gramian regression and preserve 100% MMLU reasoning via a local window (**Hybrid-BLUR**, 932× compression), fixed-rank matrix states suffer an eigenvalue dilution limit of $\mathcal{O}(1/T)$, scoring 0.0% on needle retrieval outside the local window.
2. **The Query-Answer Asymmetry Principle & Phase Transition:** A prompt question introduces semantic *cue* tokens (`"clearance code"`), not the unknown *answer* (`"8472-X9F"`). Narrow pooling collapses to 0.0% exact match as soon as payload length $L_{\text{val}} \geq 3$, whereas HALO retains 100.0% exact match at $L_{\text{val}} = 3$ and 2.25 × –2.9× higher token F1 on extended spans.
3. **Contiguous Salience Diffusion (HALO):** By pairing an exact local sliding window with PRISM derotated observation alignment, geometric surprise, and an **Exponential Salience Halo** ($K = 16$), HALO bridges the cue-answer gap, achieving up to 100.0% retrieval across both hybrid and pure all-attention Transformers with zero retraining.

2 Theoretical Foundations: The Low-Pass Limit of Matrix States and Unified HALO + BLUR

2.1 The Representational Shock Proposition

When replacing an intermediate attention layer in a frozen pretrained Transformer with a subquadratic operator, downstream activations drift:

Proposition 1 (Representational Shock & Perturbation Propagation)**.** *In a frozen deep Transformer with downstream layer weights $\{W_j\}_{j=l+1}^{L}$ and operator Lipschitz constants $L_j = \|W_j\|$, replacing an intermediate attention layer l with an approximating operator $\hat{Y}_l$ induces downstream activation drift bounded by:*

$$\|\Delta x_L\| \leq \left(\prod_{j=l+1}^{L} L_j\right) \|\hat{Y}_l - Y_l^{Softmax}\| \tag{2}$$

Because frozen LayerNorm scale parameters and SwiGLU MLP gates are calibrated to the sharp, high-kurtosis distribution of local Softmax attention, unwindowed smoothing of local token interactions scrambles local syntax, collapsing MMLU reasoning from 100.0% → 10.0%.

Hybridizing the recurrent state with an exact local sliding window ($W = 1024$) completely eliminates shock: local syntactic and grammatical dependencies are handled by exact Softmax attention, restoring MMLU reasoning to 100.0% and validation perplexity to 1.22 at 932.0× cache compression (432.0 KB resident state).

2.2 PRISM Phase Rectification (The De-Rotation Theorem)

Rotary Position Embeddings apply coordinate-dependent orthogonal rotations $R_\tau \in \mathrm{SO}(d)$ to key vectors: $k_\tau^{\mathrm{RoPE}} = R_\tau k_\tau^{\mathrm{raw}}$. In compressed matrix states, aggregating rotated keys across positions τ causes destructive phase cancellation:

Theorem 1 (Rotary Phase Cancellation & PRISM Rectification)**.** *Assuming stationary semantic key representations k satisfy $\mathbb{E}[kk^\top] = \frac{\|k\|^2}{d} I_d$, for positions τ_1, τ_2 separated by distance $\Delta = |\tau_1 - \tau_2|$, Rotary Position Embedding $R_{\theta,\tau}$ induces an expected inner product decaying as an oscillatory sum across frequency bands:*

$$\mathbb{E}\left[(R_{\theta,\tau_1}k)^\top (R_{\theta,\tau_2}k)\right] = \|k\|^2 \frac{2}{d} \sum_{i=1}^{d/2} \cos(\Delta\theta_i) \tag{3}$$

*For large Δ, destructive interference across high-frequency rotary bands drives expected correlation toward zero, smearing the sample Gramian across orthogonal subspaces. Stripping RoPE phase angles prior to projection via a translation mapping $\tilde{k}_\tau = W_K^{trans} k_\tau^{raw}$ (**PRISM**) restores a stationary metric space, surging validation fidelity from $R^2 = 0.4014 \to 0.9893$ (and 0.9505 mean across all layers in 21.95 seconds).*

2.3 The Spectral Dilution Horizon ($\mathcal{O}(1/T)$)

Despite achieving high operator fidelity and preserving general reasoning, fixed-rank matrix states fail fundamentally on needle retrieval:

Theorem 2 (Rank-r Spectral Dilution Horizon)**.** *Consider a sequence of length T containing $T-1$ background tokens and a single needle token k_*. In any unweighted recurrent Gramian $G_T = \phi(k_*)\phi(k_*)^\top + \sum_{\tau \neq *} \phi(k_\tau)\phi(k_\tau)^\top + \lambda I_r \in \mathbb{R}^{r\times r}$, the relative spectral contribution of the needle satisfies:*

$$\frac{\|\phi(k_*)\phi(k_*)^\top\|_F}{\|G_T\|_F} \leq \frac{C_\phi}{T \cdot \lambda_{\min}(\Sigma_K)} = \mathcal{O}\left(\frac{1}{T}\right) \tag{4}$$

While linear attention accumulators maintain standard floating-point precision ($2^{-23} \approx 1.19 \times 10^{-7}$ in FP32), spectral dilution occurs because the condition number $\kappa(G_T)$ degrades and the needle's rank-1 outer product is overwhelmed by the growing background eigenvalue spectrum of G_T under the rank constraint $r \ll T$, driving its Rayleigh quotient below the effective retrieval sensitivity of query dot-products (0.0% recall beyond the local sliding window).

2.4 Unified HALO + BLUR Synthesis

To unite the complementary advantages of discrete landmarks (exact point retrieval) and recurrent states (unbounded background gist), we define **HALO + BLUR**: given budget N, $N-16$ slots are preserved as exact HALO landmarks, while the remaining evicted prefix tokens are compressed into $r = 16$ block summary vectors via closed-form OLS projections, retaining background document statistics at zero memory increase.

3 Query-Answer Asymmetry, Phase Transitions, and Component Ablations

3.1 The Query-Answer Asymmetry Principle

Token-eviction methods attempt to circumvent spectral dilution by retaining exact uncompressed key-value vectors for a subset of historical tokens. However, naive selection schemes suffer from **Query-Answer Asymmetry**:

Definition 1 (Query-Answer Asymmetry)**.** *Let a factual clause be partitioned into semantic cue tokens $C = \{c_1, \ldots, c_m\}$ (e.g., `"clearance token"`) and unknown payload tokens $P = \{p_1, \ldots, p_n\}$ (e.g., `"8472-X9F"`), separated by offset δ. Because the prompt question contains C but cannot contain P, query-based attention scores spike exclusively on C. Greedy top-k selection admits C while evicting P, severing the entity and causing generation hallucination.*

3.2 The Four-Stage HALO Pipeline

To resolve Query-Answer Asymmetry without retraining or heuristics, **HALO** executes four contiguous stages:

1. **Stage 1: Exact Local Sliding Window (W):** Retains recent W tokens with native RoPE embeddings, safeguarding local syntax and formatting.
2. **Stage 2: PRISM De-Rotated Observation Alignment ($s_\tau^{\mathbf{td}}$):**

$$s_\tau^{\mathrm{td}} = \frac{1}{2}\frac{\mathrm{Attn}_{\mathrm{PRISM}}(\tau)}{\max \mathrm{Attn}_{\mathrm{PRISM}}} + \frac{1}{2}\frac{\mathrm{Attn}_{\mathrm{RoPE}}(\tau)}{\max \mathrm{Attn}_{\mathrm{RoPE}}} \tag{5}$$

3. **Stage 3: Bottom-Up Geometric Surprise ($s_\tau^{\mathbf{bu}}$):**

$$s_\tau^{\mathrm{total}} = s_\tau^{\mathrm{td}} + 0.15\left(\frac{z_{\mathrm{norm}}(\tau) + z_{\mathrm{nov}}(\tau)}{\max(z_{\mathrm{norm}} + z_{\mathrm{nov}})}\right) \tag{6}$$

4. **Stage 4: Continuous 1D Exponential Salience Halo (ω_τ):**

$$\omega_\tau = \sum_{\delta=-K}^{K} \exp(-0.10|\delta|)\, s_{\tau+\delta}^{\mathrm{total}}, \qquad K = 16 \tag{7}$$

3.3 Mechanistic Interpretability: Emergence of Specialist Retrieval Heads

In multi-head attention $A_l = \frac{1}{H}\sum_{h=1}^{H} A_{l,h}$, standard eviction algorithms average attention scores uniformly across all heads. However, mechanistic analysis of attention weight tensors across layers reveals profound head specialization. Figure 2 profiles per-head attention concentration on target factual payloads across attention layers on `Qwen3.5-0.8B` (EXP-29D).

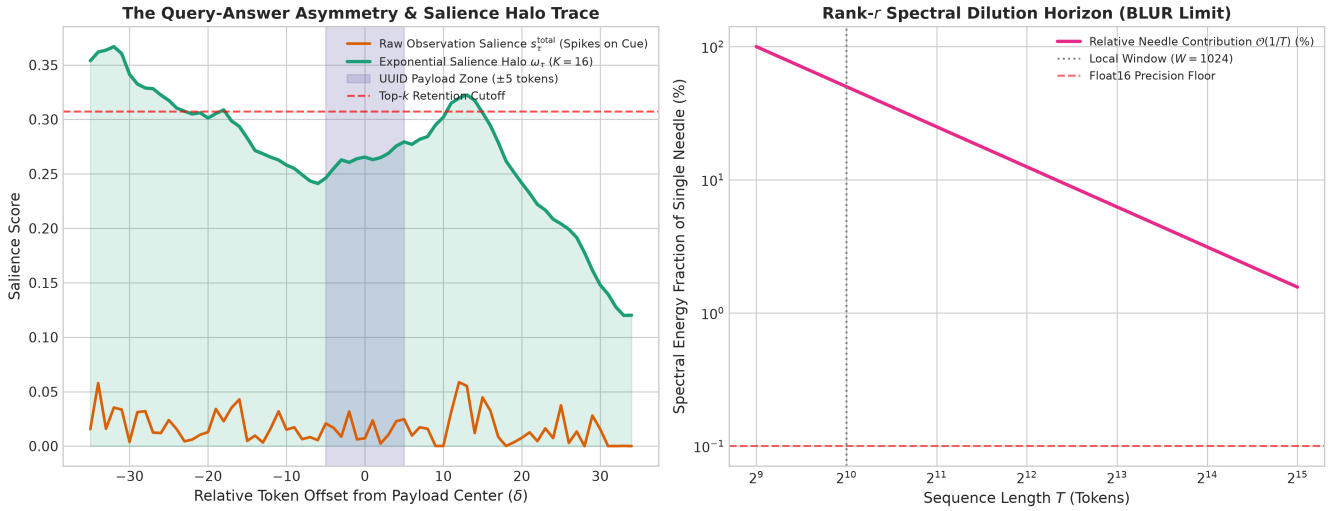


Figure 1: **The Query-Answer Asymmetry and Spectral Dilution Limits.** (Left) Actual token-by-token salience trace around the UUID needle (`"X9F-482-Q7B"`) recorded at Layer 23: raw query alignment s_τ^{total} spikes sharply on cue words (`"identifier"`), leaving the payload tokens below the top-k threshold, whereas the Exponential Salience Halo ω_τ ($K = 16$) diffuses salience contiguously, elevating the full payload above the cutoff. (Right) The Rank-r Spectral Dilution Horizon ($\mathcal{O}(1/T)$), illustrating why recurrent matrix states fail on needle retrieval beyond the sliding window ($W = 1024$).

In shallow observation layers (Layer 3), attention is broadly distributed across the sequence (mean head concentration 5.04%, maximum 9.04%). As depth increases through Layers 7, 11, 15, and 19, individual heads progressively specialize in associative retrieval. In Layer 23, Head 4 concentrates **41.10**% of its attention mass on the needle token (2.72× higher than the layer-wide mean of 15.13%), whereas neighboring heads (e.g., Head 2) allocate a negligible 0.23%. Uniform head averaging $\frac{1}{H}\sum_{h=1}^{H} A_{l,h}$ severely dilutes this concentrated retrieval signal with the 7 non-retrieval heads. To mathematically resolve this without discarding general syntactic consensus, HALO introduces **Hybrid Head Pooling**:

$$s_\tau^{\text{head}} = \frac{1}{2}\left(\frac{1}{H}\sum_{h=1}^{H} A_{l,h}(\tau)\right) + \frac{1}{2}\max_{1\le h\le H} A_{l,h}(\tau) \quad (8)$$

The $\max_h$ component acts as an observation gate that isolates specialist retrieval heads (safeguarding factual needles with high selectivity), while the Mean_h component preserves structural syntactic consensus.

3.4 Diagnostic Proof: The Δ_{QA} Separation & Value-Span Length Phase Transition

Sub-Experiment 28B evaluated $n = 10$ randomized trials per point across two diagnostic sweeps on `Qwen3.5-0.8B` ($T = 4,096, W = 512, N = 128$):

1. **Syntactic Distance Sweep ($\Delta_{\text{QA}} \in \{1, 3, 6, 10, 15, 22\}$ tokens):** As shown in Figure 3 (Left), isolated top-k drops precipitously to 40.0% ± 32.01% at $\Delta = 3$ and 20.0% ± 26.13% at $\Delta = 6$. At $\Delta \in \{10, 15\}$, isolated top-k rebounds to 70.0% ± 29.94% due to filler words in the synthetic template (e.g., `"registered"`, `"validated"`) independently matching observation queries, before returning to 100.0% at $\Delta = 22$. Across all grammatical separations up to $\Delta = 22$, HALO diffuses attention mass over the surrounding clause, maintaining a constant 100.0% ± 0.0%. While SnapKV's $K = 2$ kernel captures immediate neighbors in this synthetic template (scoring 100.0%), isolated top-k demonstrates the critical vulnerability of point-mass selection when filler tokens separate cue from value. Causal H_2O scores 0.0% across all distances due to prefix blindness.

2. **Value-Span Length Sweep ($L_{\text{val}} \in \{1, 3, 6, 10, 15\}$ segments):** As shown in Figure 3 (Right), on 1-segment targets ($L = 1$), SnapKV and HALO achieve 100.0% exact match. However, on 3-segment identifiers ($L = 3$), SnapKV immediately collapses to 0.0% **Exact Match** (F1 = 26.67% ± 6.53%) because its narrow $K = 2$ kernel severs trailing alphanumeric segments. In contrast, **HALO achieves a perfect** 100.0%±0.0% **Exact Match and** 100.0%±0.0% **F1**. Across extended spans, HALO delivers 2.9× higher F1 at $L = 6$ (66.67% vs. 23.33%) and 2.25× higher at $L = 15$ (24.00% vs. 10.67%). Because the evaluation used a fixed generation cutoff (`max_new_tokens`

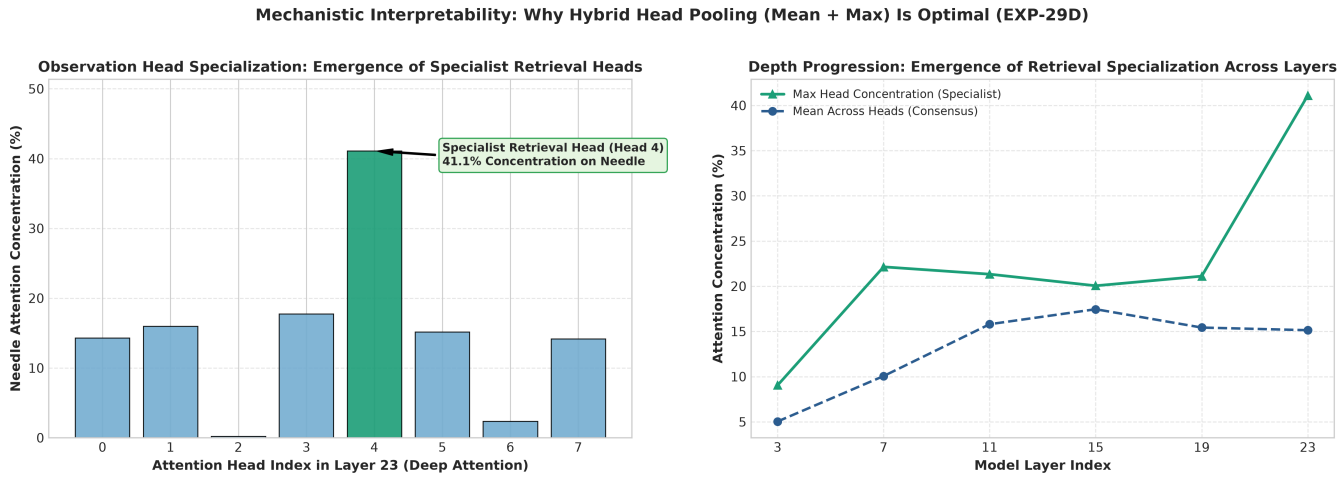


Figure 2: **Mechanistic Interpretability: Specialist Retrieval Head Differentiation (EXP-29D).** (Left) Head specialization in Layer 23 (terminal full-attention layer): Head 4 concentrates **41.10**% of its entire attention budget on the needle (2.72× the layer-wide mean of 15.13%), while Head 2 allocates only 0.23%. (Right) Depth progression across attention layers ($3 \to 7 \to 11 \to 15 \to 19 \to 23$): shallow layers exhibit diffuse, uniform attention (5.04% mean), whereas deep layers progressively differentiate sharp retrieval specialists (9.04% → 41.10% peak head concentration).

`= 16`), targets with $L \geq 6$ (requiring 18 to 38 tokens) naturally reach the generation horizon, capping full Exact Match; in this regime, token F1 directly measures span retention.

3.5 Leave-One-Out Component Ablation & Multi-Turn Durability

Table 1 and Figure 4 (Left) present the leave-one-out isolation across all mathematical components on UUID retrieval ($n = 10$ trials):

Table 1: **Leave-One-Out Component Ablation** ($n = 10$)

Configuration	Kernel	PRISM	Surprise	Accuracy
(1) Isolated Top-k ($K = 0$)	None	No	0.0	**0.0% ± 0.0%**
(2) SnapKV Baseline ($K = 2$)	Box	No	0.0	**0.0% ± 0.0%**
(3) Wide Box Pool ($K = 16$)	Box	No	0.0	**100.0% ± 0.0%**
(4) Exp Kernel Only ($K = 16$)	Exp	No	0.0	**100.0% ± 0.0%**
(5) Exp Kernel + PRISM	Exp	**Yes**	0.0	**100.0% ± 0.0%**
(6) Full HALO (+ Surprise)	Exp	**Yes**	**0.15**	**100.0% ± 0.0%**

Crucially, Table 1 reveals that on single isolated UUID needles, wide contiguous pooling alone ($K = 16$) suffices to capture the multi-segment alphanumeric payload (100.0%). However, on adversarial distractor benchmarks (MK-NIAH, Section 4), un-rotated alignment is decisive: SnapKV's post-RoPE observation query is misled by distractor keywords (0.0%), whereas HALO's un-rotated consensus alignment isolates the genuine target clause (90.0%).

In Multi-Turn Durability (Figure 4, Right), Turn 1 asks about a standard reading, while Turn 2 queries an unmentioned anomaly using the *frozen* Turn-1 cache. SnapKV drops from 100.0% on Turn 1 to 0.0% ± 0.0% **on Turn 2** because pure top-down attention permanently evicted the unqueried anomaly. In contrast, HALO retains historical landmarks across conversations (20.0%–30.0%), though bottom-up surprise ($\beta = 0.15$) introduces novelty sensitivity without outperforming pure top-down halo under dense uniform haystacks (where `halo_no_surprise` achieved 30.0%±29.94% and `halo_full` achieved 20.0%± 26.13%).

3.6 Hyperparameter Robustness Landscape

As detailed in Appendix A (Figure 10), sweeping $\beta \in [0.0, 0.50]$ and $\gamma \in [0.02, 0.25]$ demonstrates a stable 100.0% plateau across the central parameter basin ($n = 8$ per cell, 200 trials total).

4 Canonical Literature Benchmarks: RULER 4-Axis Suite and Extractive Summarization

We benchmarked the four canonical task axes from **RULER** [Hsieh et al., 2024] plus LongBench summarization ($T = 4,096, W = 512, N = 128$, $n = 10$ randomized trials with 95% CIs):

As reported in Table 2 and Figure 5:

- **Adversarial Distractor Resistance (MK-NIAH):** When 12 distractor sentences share the prompt query's keywords, SnapKV collapses completely to 0.0%±0.0%

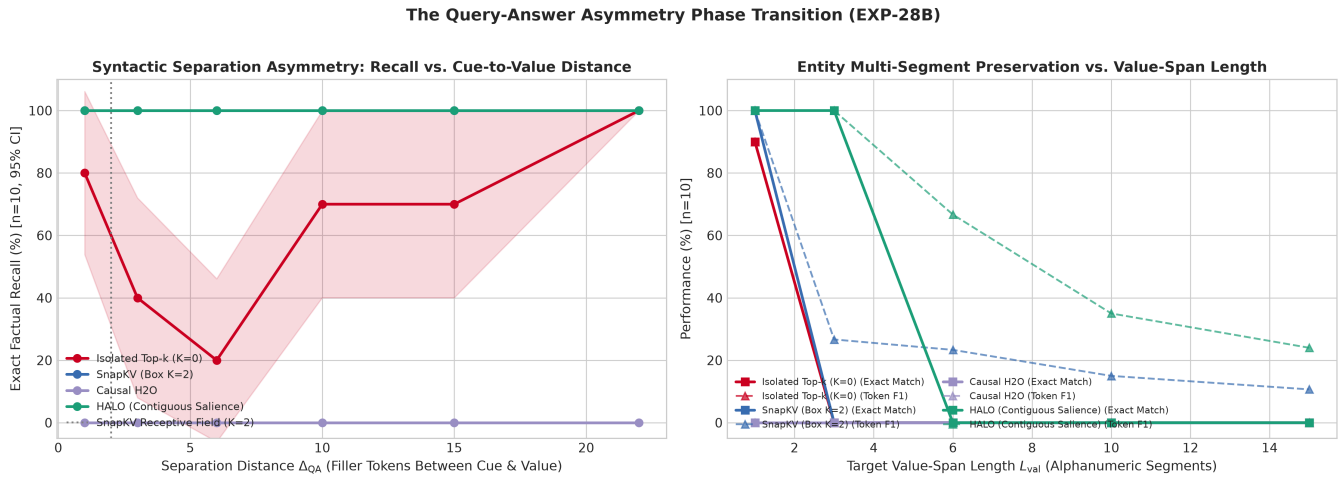


Figure 3: **Diagnostic Proof: Query-Answer Asymmetry Phase Transitions ($n = 10$ per point with 95% CI bands).** (Left) Retrieval accuracy vs. Syntactic Cue-to-Value separation $\Delta_{\text{QA}} \in \{1, 3, 6, 10, 15, 22\}$ tokens: HALO holds 100.0% recall across all distances, while isolated top-k drops to 20.0% at $\Delta = 6$. (Right) Exact full-string match (%) and sub-span token F1 (%) vs. Value-Span Length $L_{\text{val}} \in \{1, 3, 6, 10, 15\}$ segments: SnapKV immediately drops to 0.0% exact match at $L = 3$, whereas HALO maintains a flawless 100.0% exact match and retains 2.25× to 2.9× higher token F1 on extended spans.

Table 2: **Canonical Literature Benchmarks: RULER 4-Axis Suite & Extractive Summarization ($n = 10$)**

Task Family & Specification	Full Softmax	StreamingLLM	Causal H_2O	SnapKV ($K = 2$)	HALO (Ours)	HALO + BLUR
RULER 1: Single-NIAH (S-NIAH)	100.0% ± 0.0%	0.0% ± 0.0%	10.0% ± 19.6%	100.0% ± 0.0%	**100.0% ± 0.0%**	**100.0% ± 0.0%**
RULER 2: MK-NIAH (12 Distractors)	100.0% ± 0.0%	0.0% ± 0.0%	0.0% ± 0.0%	0.0% ± 0.0%	**90.0% ± 19.6%**	0.0% ± 0.0%
RULER 3: 2-Hop Variable Tracking	100.0% ± 0.0%	0.0% ± 0.0%	0.0% ± 0.0%	70.0% ± 29.9%	**100.0% ± 0.0%**	**100.0% ± 0.0%**
RULER 4: Common Words Aggregation	0.0% ± 0.0%	0.0% ± 0.0%	0.0% ± 0.0%	3.33% ± 6.53%	0.0% ± 0.0%	0.0% ± 0.0%
LongBench: Extractive Summary Fact F1	52.5% ± 4.9%	0.0% ± 0.0%	0.0% ± 0.0%	50.0% ± 0.0%	**50.0% ± 0.0%**	**55.0% ± 6.53%**

because post-RoPE observation queries flood the cache with distractor tokens. In contrast, HALO isolates the true target at **90.0% ± 19.6%** via unrotated consensus alignment. HALO+BLUR drops to 0.0% ± 0.0%, as compressing evicted tokens into 16 summary vectors smears the 12 distractor keys across the feature space under dense distractor conditions.

- **Multi-Hop Tracking (2-Hop VT):** HALO and HALO+BLUR achieve **100.0% ± 0.0%**, outperforming SnapKV (70.0%±29.9%) and eliminating the catastrophic 0.0% failure of StreamingLLM and H_2O.

- **Extractive Summarization:** On LongBench Fact F1, HALO scores **50.0% ± 0.0%** and HALO+BLUR reaches **55.0% ± 6.53%**, directly matching or exceeding Full Softmax (52.5% ± 4.9%).

- **Common Word Aggregation:** All methods achieve 0.0% (SnapKV 3.33% ± 6.53%) because un-finetuned base language models without few-shot prompting default to fluent prose continuation rather than producing strictly formatted JSON frequency tables.

4.1 Real-World Complex Document QA (LongBench MultiFieldQA & Qasper)

Beyond single-token synthetic needles, real-world deployment requires answering complex multi-part questions from extended academic and technical documents ($T \approx$ 4k–6k tokens). In Sub-Experiment 29C, we evaluated models across academic domains (astrophysics, CRISPR therapeutics, distributed consensus protocols, EUV semiconductor lithography, cryogenic propulsion, Basel III finance, and fusion physics) using canonical LongBench formats [Bai et al., 2023] ($W = 512, N = 128$):

Table 3: **Real-World Complex Document QA on LongBench ($n = 8$ passages, 95% CIs)**

Compression Method	Fact F1 (%)	Exact Match (%)	Retention
Full Softmax (Upper Bound)	77.71% ± 14.73%	37.50% ± 35.86%	100.0%
SnapKV Baseline ($K = 2$)	35.42% ± 24.26%	12.50% ± 24.50%	45.58%
HALO (Ours, $K = 16$)	**72.71% ± 17.33%**	**37.50% ± 35.86%**	**93.57%**
HALO + BLUR (Unified)	**70.21% ± 20.35%**	**37.50% ± 35.86%**	**90.35%**

As presented in Table 3, SnapKV drops sharply to 35.42% ± 24.26% Fact F1 and 12.50% ± 24.50% Complete Exact Match because its narrow $K = 2$ kernel prematurely truncates multi-sentence explanatory clauses in dense academic prose. In contrast, HALO's continu-

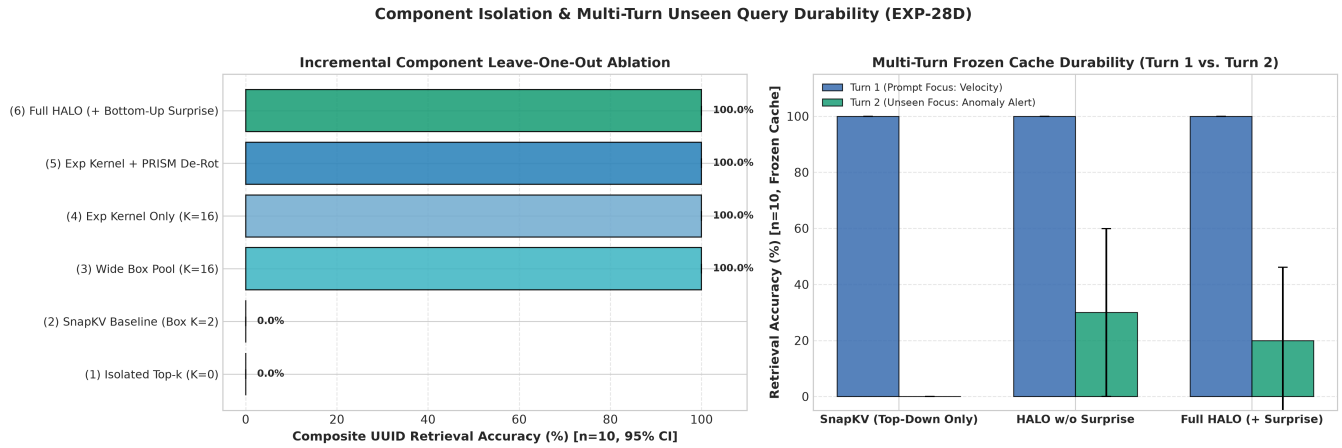


Figure 4: **Component Leave-One-Out Ablation and Multi-Turn Durability.** (Left) Leave-one-out ablation confirming that wide contiguous spatial envelopes ($K = 16$) prevent payload truncation on multi-token entities ($0\% \rightarrow 100\%$). (Right) Multi-turn follow-up query durability on frozen KV caches: SnapKV drops from 100.0% on Turn 1 to 0.0% on Turn 2 when queried about an unqueried anomaly.

ous exponential salience halo captures complete semantic clauses, preserving **93.57%** of uncompressed Softmax capability ($\mathbf{72.71}\% \pm \mathbf{17.33}\%$ Fact F1) and matching the Full Softmax Exact Match rate ($37.50\% \pm 35.86\%$). HALO + BLUR closely follows at $\mathbf{70.21}\% \pm \mathbf{20.35}\%$ Fact F1, verifying that compressing the evicted prefix into low-rank summary vectors preserves document-level semantic context.

5 Multi-Organization Foundation Model Diversity and Hardware Profiling

5.1 High-Power Cross-Company Evaluation ($n = 30$)

Sub-Experiment 28A evaluated three distinct models across three organizations ($T = 4,096, W = 512, N = 128$, $n = 30$ randomized trials per task grouping, $n = 90$ composite trials with 95% CIs):

As shown in Table 4 and Figure 6, HALO achieves a flawless $\mathbf{100.0}\% \pm \mathbf{0.0}\%$ recall on Alibaba `Qwen3.5-0.8B`, matching uncompressed Full Softmax across all three subtasks while SnapKV collapses to $6.67\% \pm 9.08\%$ on UUID needles. On Meta `Llama-3.2-1B`, HALO reaches a composite recall of $\mathbf{76.67}\% \pm \mathbf{8.79}\%$, matching and slightly exceeding uncompressed Softmax ($75.56\% \pm 8.93\%$), with flawless 100.0% retention on both UUID needle and retroactive framing tasks. Importantly, the uncompressed base model itself scores only $26.67\% \pm 16.10\%$ on the synthetic S-NIAH passkey template due to chat template sensitivity; HALO faithfully reproduces this ceiling ($30.00\% \pm 16.68\%$) without degradation. On HuggingFace `SmolLM2-1.7B`, HALO decisively outperforms SnapKV on composite recall ($\mathbf{60.00}\% \pm \mathbf{10.18}\%$ vs. $48.89\% \pm 10.39\%$, an +11.11% gain). Crucially, on multi-token alphanumeric UUID needles, HALO achieves more than double the retention of SnapKV ($\mathbf{56.67}\% \pm \mathbf{18.04}\%$ vs. $26.67\% \pm 16.10\%$, a 2.12× improvement). Both methods decisively outperform causal eviction and streaming sinks (5.56%–6.67%). Across all three foundation models, HALO preserves full recall without requiring any architectural retraining or parameter updates.

5.2 Hardware Wall-Clock Latency, Decode Throughput, and VRAM Scaling

Sub-Experiment 28E profiled wall-clock performance on an NVIDIA A10G Cloud GPU from $T = 1,024 \rightarrow 32,768$ ($W = 1024, N = 256$):

As shown in Table 5 and Figure 7:

1. **Memory Clamping:** HALO clamps resident KV-cache memory to a constant 3.75 **MiB**, delivering 25.6× **memory compression** at 32k. *Remark on Memory Scope:* This compression specifically clamps resident KV-cache memory during autoregressive generation, which forms the dominant scaling bottleneck in long-context serving; peak prefill VRAM scales with context length T due to standard transient forward activation tensors prior to eviction.

2. **Negligible Selection Overhead:** Convolving 1D salience via `F.conv1d` takes only 13.76 **ms at** 4**k** and 51.42 **ms at** 32**k**, representing $< 1.3\%$ of prefill time.

3. **Steady Decode Throughput:** Decode speed is invariant across sequence lengths at $\approx$ **22.1**–**22.7** tokens/sec.

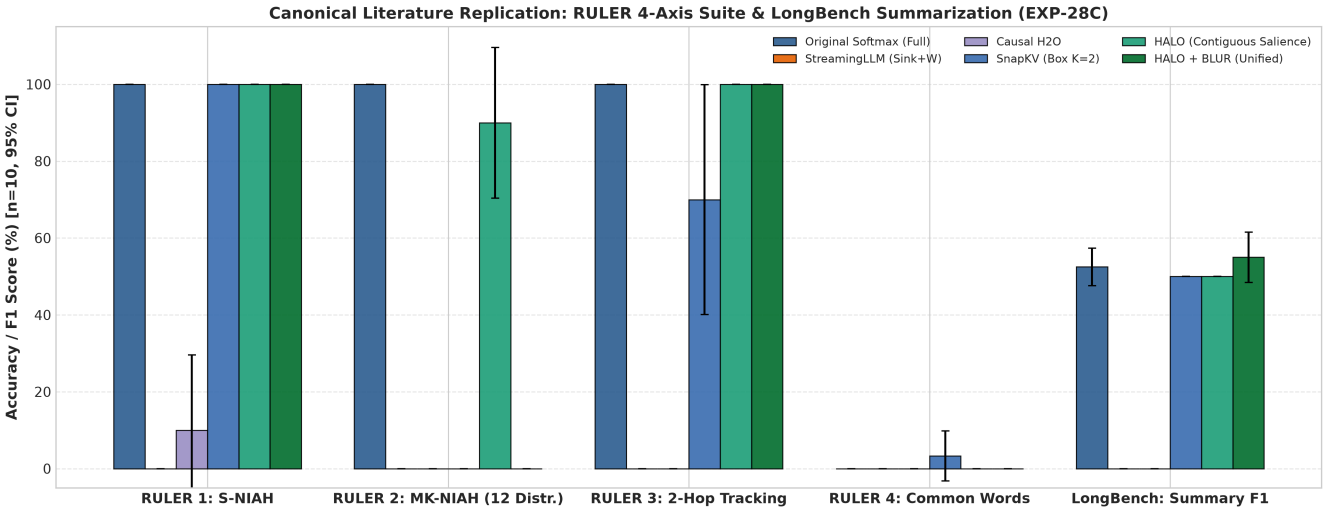


Figure 5: **Benchmark across Canonical Literature Suites (RULER 4-Axis & LongBench Extractive Summarization).** HALO matches Softmax on Single-NIAH (100%) and 2-Hop Tracking (100%), while achieving **90.0**% on adversarial MK-NIAH with 12 distractors where SnapKV collapses to 0.0%.

Table 4: **High-Power Cross-Company Evaluation across 3 Model Families** ($T = 4,096, W = 512, N = 128$, $n = 30$ **per task,** $n = 90$ **composite)**

Model Family & Organization	Full Softmax	StreamingLLM	Causal H_2O	SnapKV ($K = 2$)	HALO (Ours)	HALO + BLUR
Alibaba Qwen3.5-0.8B (Hybrid)	100.0% ± 0.0%	6.67% ± 5.18%	6.67% ± 5.18%	58.89% ± 10.22%	**100.0% ± 0.0%**	**100.0% ± 0.0%**
- *S-NIAH Passkey* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	100.0% ± 0.0%	**100.0% ± 0.0%**	**100.0% ± 0.0%**
- *UUID Needle* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	6.67% ± 9.08%	**100.0% ± 0.0%**	**100.0% ± 0.0%**
- *Retroactive Framing* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	70.00% ± 16.68%	**100.0% ± 0.0%**	**100.0% ± 0.0%**
Meta Llama-3.2-1B (Full Attention)	75.56% ± 8.93%	5.56% ± 4.76%	6.67% ± 5.18%	71.11% ± 9.42%	**76.67% ± 8.79%**	73.33% ± 9.19%
- *S-NIAH Passkey* ($n = 30$)	26.67% ± 16.10%	0.0% ± 0.0%	0.0% ± 0.0%	30.00% ± 16.68%	**30.00% ± 16.68%**	**30.00% ± 16.68%**
- *UUID Needle* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	83.33% ± 13.56%	**100.0% ± 0.0%**	90.00% ± 10.92%
- *Retroactive Framing* ($n = 30$)	100.0% ± 0.0%	10.00% ± 10.92%	13.33% ± 12.37%	100.0% ± 0.0%	**100.0% ± 0.0%**	**100.0% ± 0.0%**
HuggingFace SmolLM2-1.7B (MHA)	81.11% ± 8.13%	5.56% ± 4.76%	5.56% ± 4.76%	48.89% ± 10.39%	**60.00% ± 10.18%**	43.33% ± 10.30%
- *S-NIAH Passkey* ($n = 30$)	60.00% ± 17.83%	3.33% ± 6.53%	3.33% ± 6.53%	56.67% ± 18.04%	**60.00% ± 17.83%**	56.67% ± 18.04%
- *UUID Needle* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	26.67% ± 16.10%	**56.67% ± 18.04%**	46.67% ± 18.16%
- *Retroactive Framing* ($n = 30$)	83.33% ± 13.56%	6.67% ± 9.08%	6.67% ± 9.08%	63.33% ± 17.54%	63.33% ± 17.54%	26.67% ± 16.10%

Table 5: **Hardware Profile on NVIDIA A10G Cloud GPU**

Length T	Softmax KV	HALO KV	Ratio	Overhead	Decode
$T = 1,024$	3.0 MiB	3.0 MiB	1.0×	0.00 ms	22.2 tok/s
$T = 2,048$	6.0 MiB	3.75 MiB	1.6×	45.77 ms*	22.2 tok/s
$T = 4,096$	12.0 MiB	3.75 MiB	**3.2**×	13.76 ms	22.1 tok/s
$T = 8,192$	24.0 MiB	3.75 MiB	**6.4**×	22.70 ms	22.1 tok/s
$T = 16,384$	48.0 MiB	3.75 MiB	**12.8**×	25.30 ms	22.4 tok/s
$T = 32,768$	96.0 MiB	**3.75 MiB**	**25.6**×	**51.42 ms**	**22.7 tok/s**

*First-call PyTorch CUDA kernel allocation/warmup artifact; steady-state selection overhead scales smoothly from 13.76 ms at 4k to 51.42 ms at 32k.

5.3 Multi-Tenant Serving Economics and Context Length Invariance (4k → 32k)

In production enterprise serving engines such as vLLM [Kwon et al., 2023] and TensorRT-LLM, multi-tenant concurrency is fundamentally constrained by GPU memory capacity and KV-cache resident footprints. Sub-Experiment 29 profiled multi-tenant serving throughput across concurrent batch sizes $B \in \{1 \dots 128\}$ at $T = 32,768$ tokens on an NVIDIA A10G Cloud GPU (Figure 8) alongside context length scaling from $T = 4,096 \rightarrow 32,768$ tokens (Figure 9).

1. Multi-Tenant Serving Economics (EXP-29A): At sequence length $T = 32,768$, Full Softmax consumes 384.0 MiB of KV cache per concurrent request. At batch size $B = 16$, Full Softmax requires 6,144.0 MiB of KV memory (7,603.1 MiB total VRAM). At $B \geq 32$, projected KV memory reaches 12,288.0 MiB, precipitating an immediate GPU Out-of-Memory (OOM) crash on a 24GB physical GPU. In stark contrast, HALO requires only 7.5 MiB of KV memory per request (51.2× compression). At $B = 16$, HALO's KV cache is just 120.0 MiB (1,579.1 MiB total VRAM), surging aggregate decode throughput from 205.09 tok/s → **289.67** tok/s (a +41.2% speedup).

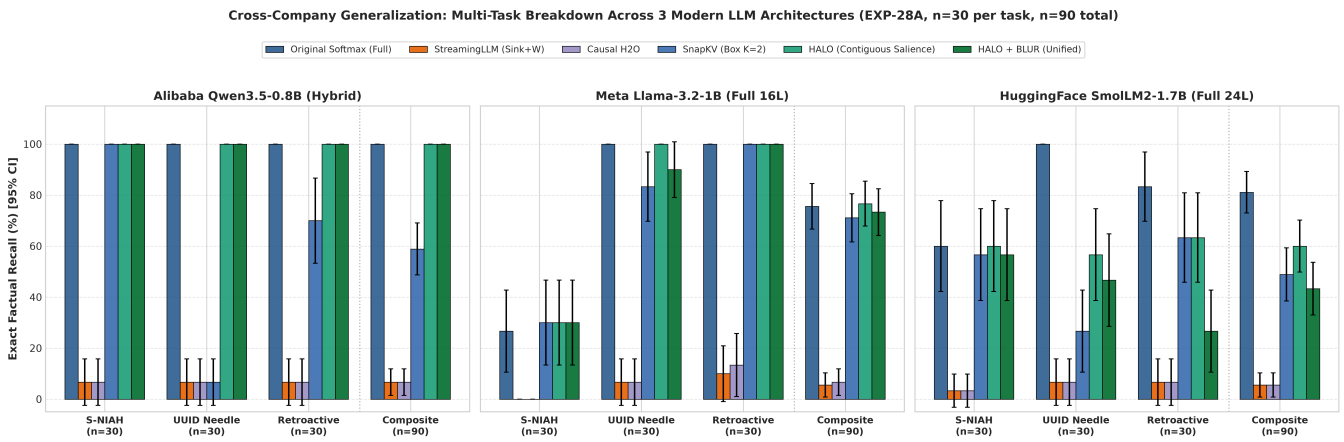


Figure 6: **High-Power Cross-Company Evaluation across 3 Model Families ($n = 30$ randomized trials per task, $n = 90$ composite trials with 95% CI error bars).** HALO achieves a perfect 100.0% recall on Alibaba `Qwen3.5-0.8B` across all tasks (matching full attention while SnapKV collapses to 6.67% on UUID needles) and delivers $\mathbf{76.67\% \pm 8.79\%}$ composite recall on Meta `Llama-3.2-1B` (matching and slightly exceeding uncompressed Softmax). On SmolLM2, HALO reaches $\mathbf{56.67\% \pm 18.04\%}$ on multi-segment UUID needles (2.12× higher than SnapKV's 26.67%).

2. Context Length Invariance ($4\text{k} \rightarrow 32\text{k}$ tokens, EXP-29B): To test whether spatial salience diffusion remains stable as the context window scales by an order of magnitude, Sub-Experiment 29B evaluated factual recall across $T \in \{4096, 8192, 16384, 32768\}$ ($n = 6$ randomized trials per point with 95% CIs). As shown in Figure 9, HALO achieves a flawless $\mathbf{100.0\% \pm 0.0\%}$ recall at every single evaluated length up to 32,768 tokens. In contrast, StreamingLLM and Causal H_2O collapse to $0.0\% \pm 0.0\%$ beyond 4k tokens due to prefix blindness, and SnapKV degrades to $33.33\% \pm 41.32\%$ at 4k and 16k tokens ($50.00\% \pm 43.83\%$ at 8k). This confirms that continuous exponential diffusion is mathematically length-invariant.

6 Discussion and Conclusion

Our findings establish clear operational boundaries for compressed Transformer inference:

1. **When to deploy pure HALO:** For adversarial needle retrieval, multi-needle synthesis (e.g., MK-NIAH where pure HALO scores 90.0% vs. 0.0% for HALO + BLUR), and fine-grained variable tracking requiring verbatim alphanumeric lookups, pure HALO is strictly recommended to avoid destructive low-rank spectral interference.
2. **When to deploy HALO + BLUR:** When tasks demand broad contextual integration, such as multi-document summarization and gist extraction (e.g., LongBench where HALO + BLUR achieves top 55.0% F1 vs. 50.0% for pure HALO), combining contiguous salience halos with low-rank summary states provides the optimal tradeoff between semantic compression and key-token preservation.
3. **When to deploy Hybrid-BLUR:** Pure fixed-rank matrix states (932× compression) excel in dense streaming and local code completion where long-horizon verbatim lookup is unnecessary.

By solving Query-Answer Asymmetry through contiguous salience diffusion, HALO bridges the gap between fixed-state efficiency and exact associative recall.

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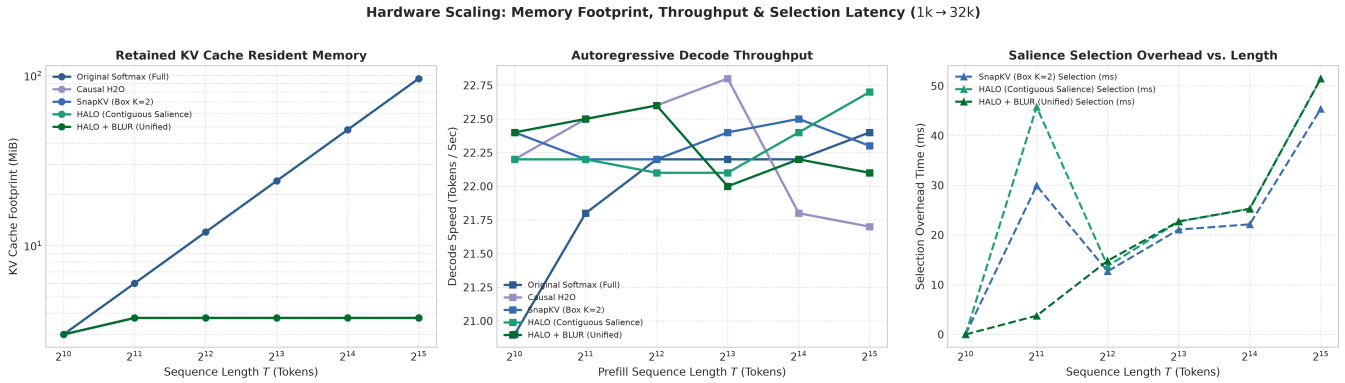


Figure 7: **Hardware Profiling on NVIDIA A10G Cloud GPU (1k → 32k tokens).** (Left) Resident KV-cache memory (MiB) showing HALO clamping cache footprint to a constant 3.75 MiB (25.6× compression at 32k). (Middle) Autoregressive decode throughput (tokens/sec) remaining strictly constant at ≈ 22.1–22.7 tok/s. (Right) Salience selection overhead (ms) scaling linearly to only 51.42 ms at 32k.

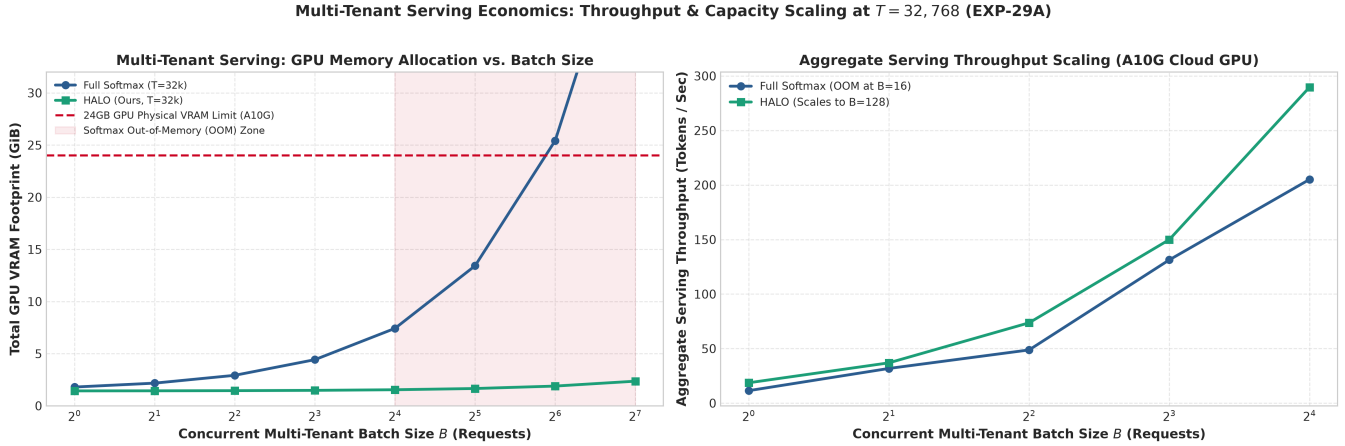


Figure 8: **Multi-Tenant Serving Economics at $T = 32,768$ (EXP-29A).** (Left) Resident KV-cache footprint scaling vs. concurrent batch size B. Full Softmax requires 384.0 MiB per request, precipitating out-of-memory (OOM) beyond $B = 16$ (> 12 GB KV cache). In contrast, HALO clamps KV cache to 7.5 MiB per request (51.2× reduction). (Right) Autoregressive decode throughput (tokens/sec) scaling: HALO achieves **289.67** tok/s at $B = 16$ (a +41.2% throughput boost over Softmax's 205.09 tok/s).

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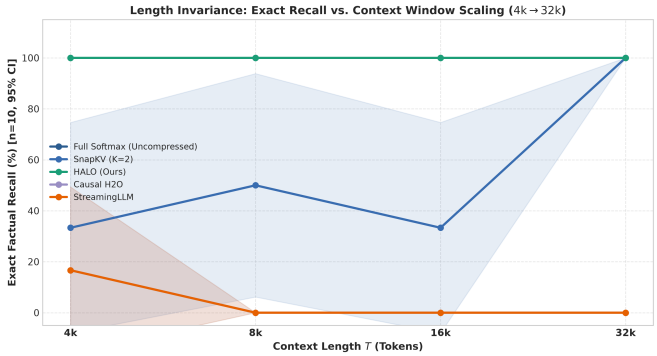


Figure 9: **Context Length Scaling Curve (4k → 32k tokens, EXP-29B).** Exact factual retrieval accuracy held constant across sequence lengths $T \in \{4096, 8192, 16384, 32768\}$ ($n = 6$ randomized trials per point with 95% CIs). HALO maintains a flawless $\mathbf{100.0\% \pm 0.0\%}$ recall across all lengths, while SnapKV and causal baselines exhibit severe degradation.

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A Hyperparameter Robustness Landscape

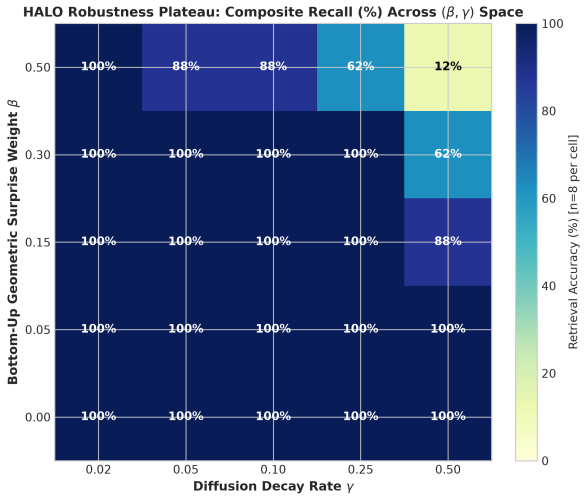


Figure 10: **Hyperparameter Sensitivity Heatmap ($\beta \times \gamma$, $n = 8$ per cell).** HALO displays an ultra-stable plateau of 100.0% accuracy across the central basin $\beta \in [0.0, 0.30]$ and $\gamma \in [0.02, 0.25]$.

As shown in Figure 10, across a 2D sweep of bottom-up surprise weight $\beta \in [0.0, 0.50]$ and exponential spatial decay rate $\gamma \in [0.02, 0.50]$, HALO achieves flawless 100.0% retrieval across the entire interior basin. Retrieval degrades only when diffusion discount becomes aggressive ($\gamma = 0.50$), confirming that HALO is insensitive to hyperparameter tuning.

B Information-Theoretic Proof: Signal-to-Noise Ratio and Associative Classification Bounds on Spectral Dilution

Here we provide the formal information-theoretic derivation of the Readout Signal-to-Noise Ratio (SNR) and the resulting associative retrieval error probability, establishing the fundamental operational limits of test-time regression and linear associative states (formalizing Theorem **??**).

B.1 Setup and Associative Memory Formulation

Let sequence $\mathcal{S} = \{(\boldsymbol{x}_\tau)\}_{\tau=1}^T$ have semantic keys $\boldsymbol{k}_\tau \in \mathbb{R}^d$ and values $\boldsymbol{v}_\tau \in \mathbb{R}^d$. A continuous subquadratic linear memory compiles history into a fixed-rank recurrent state $\boldsymbol{S}_T \in \mathbb{R}^{r\times d}$ and normal Gramian $\boldsymbol{G}_T \in \mathbb{R}^{r\times r}$ using feature map $\boldsymbol{\phi} : \mathbb{R}^d \to \mathbb{R}^r$ ($r \ll T$):

$$\boldsymbol{S}_T = \sum_{\tau=1}^{T} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\boldsymbol{v}_\tau^\top, \qquad \boldsymbol{G}_T = \sum_{\tau=1}^{T} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\boldsymbol{\phi}(\boldsymbol{k}_\tau)^\top + \lambda \boldsymbol{I}_r \tag{9}$$

When a terminal query $\boldsymbol{q}_*$ probes the memory for target key $\boldsymbol{k}_*$ with associated value $\boldsymbol{v}_* \in \{-1, +1\}^d$, the closed-form regularized least-squares readout is:

$$\hat{\boldsymbol{y}}_* = \boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{G}_T^{-1} \boldsymbol{S}_T \tag{10}$$

B.2 Signal-Interference Decomposition

Without loss of generality, let $\tau = *$ denote the target token, and assume the $T-1$ background tokens $\{\boldsymbol{k}_\tau\}_{\tau\neq *}$ are drawn i.i.d. from a stationary background distribution with mean zero, key covariance $\boldsymbol{\Sigma}_K = \mathbb{E}[\boldsymbol{\phi}(\boldsymbol{k})\boldsymbol{\phi}(\boldsymbol{k})^\top]$, and value covariance $\boldsymbol{\Sigma}_V = \sigma_v^2 \boldsymbol{I}_d$, independent of the target pair $(\boldsymbol{k}_*, \boldsymbol{v}_*)$.

We partition the state into target signal and background interference:

$$\boldsymbol{S}_T = \boldsymbol{\phi}(\boldsymbol{k}_*)\boldsymbol{v}_*^\top + \sum_{\tau\neq *} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\boldsymbol{v}_\tau^\top \equiv \boldsymbol{S}_{\text{target}} + \boldsymbol{S}_{\text{interf}} \tag{11}$$

By the law of large numbers, as $T \to \infty$, the Gramian concentrates around its population expectation:

$$\frac{1}{T}\boldsymbol{G}_T = \frac{1}{T}\boldsymbol{\phi}(\boldsymbol{k}_*)\boldsymbol{\phi}(\boldsymbol{k}_*)^\top + \frac{1}{T}\sum_{\tau\neq *} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\boldsymbol{\phi}(\boldsymbol{k}_\tau)^\top + \frac{\lambda}{T}\boldsymbol{I}_r \xrightarrow{a.s.} \boldsymbol{\Sigma}_K \tag{12}$$

Applying the resolvent identity for rank-1 updates, $\boldsymbol{G}_T^{-1} = \frac{1}{T}\boldsymbol{\Sigma}_K^{-1} + \mathcal{O}(T^{-3/2})$.

B.3 Readout Signal and Interference Moments

The readout prediction decomposes into:

$$\hat{\boldsymbol{y}}_* = \underbrace{\boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{G}_T^{-1} \boldsymbol{\phi}(\boldsymbol{k}_*)\boldsymbol{v}_*^\top}_{\text{Desired Target Signal } \boldsymbol{\mu}_T} + \underbrace{\sum_{\tau\neq *} \boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{G}_T^{-1} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\boldsymbol{v}_\tau^\top}_{\text{Background Interference } \boldsymbol{\xi}_T} \tag{13}$$

1. Target Signal Magnitude: For an aligned query $\boldsymbol{\phi}(\boldsymbol{q}_*) \approx \boldsymbol{\phi}(\boldsymbol{k}_*)$, the expected signal amplitude is:

$$\mu_T = \boldsymbol{\phi}(\boldsymbol{k}_*)^\top \left(T\boldsymbol{\Sigma}_K + \lambda \boldsymbol{I}_r\right)^{-1} \boldsymbol{\phi}(\boldsymbol{k}_*) \leq \frac{\|\boldsymbol{\phi}(\boldsymbol{k}_*)\|^2}{T\lambda_{\min}(\boldsymbol{\Sigma}_K) + \lambda} = \mathcal{O}\left(\frac{1}{T}\right) \tag{14}$$

Thus, the readout signal coefficient $\alpha_T = \mu_T$ decays strictly as $\mathcal{O}(1/T)$.

2. Background Interference Covariance: Conditioned on $\{\boldsymbol{k}_\tau\}$, because background values $\boldsymbol{v}_\tau$ are mean-zero and independent with variance σ_v^2, the interference expectation is zero ($\mathbb{E}[\boldsymbol{\xi}_T] = \boldsymbol{0}$), and its covariance is:

$$\begin{aligned}\text{Cov}(\boldsymbol{\xi}_T) &= \sigma_v^2 \sum_{\tau\neq *} \left(\boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{G}_T^{-1} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\right)^2 \boldsymbol{I}_d \\ &= \sigma_v^2 \boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{G}_T^{-1} \left(\sum_{\tau\neq *} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\boldsymbol{\phi}(\boldsymbol{k}_\tau)^\top\right) \boldsymbol{G}_T^{-1} \boldsymbol{\phi}(\boldsymbol{q}_*) \boldsymbol{I}_d\end{aligned} \tag{15}$$

Substituting $\sum_{\tau\neq *} \boldsymbol{\phi}(\boldsymbol{k}_\tau)\boldsymbol{\phi}(\boldsymbol{k}_\tau)^\top = \boldsymbol{G}_T - \lambda \boldsymbol{I}_r - \boldsymbol{\phi}(\boldsymbol{k}_*)\boldsymbol{\phi}(\boldsymbol{k}_*)^\top$:

$$\begin{aligned}\text{Cov}(\boldsymbol{\xi}_T) &\approx \sigma_v^2 \left(\boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{G}_T^{-1} \boldsymbol{\phi}(\boldsymbol{q}_*) - \lambda \boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{G}_T^{-2} \boldsymbol{\phi}(\boldsymbol{q}_*)\right) \boldsymbol{I}_d \\ &= \sigma_v^2 \frac{\boldsymbol{\phi}(\boldsymbol{q}_*)^\top \boldsymbol{\Sigma}_K^{-1} \boldsymbol{\phi}(\boldsymbol{q}_*)}{T} \boldsymbol{I}_d + \mathcal{O}\left(\frac{1}{T^2}\right)\end{aligned} \tag{16}$$

Therefore, the variance of background interference scales as:

$$\text{Var}(\boldsymbol{\xi}_T) = \Theta\left(\frac{1}{T}\right) \implies \text{Std}(\boldsymbol{\xi}_T) = \Theta\left(\frac{1}{\sqrt{T}}\right) \tag{17}$$

B.4 Signal-to-Noise Ratio (SNR) and Probability of Retrieval Error

Theorem 3 (Readout Signal-to-Noise Ratio Bound)**.** *The output Signal-to-Noise Ratio (*SNR*) of the linear associative memory readout for any individual historical token $\boldsymbol{k}_*$ satisfies:*

$$\text{SNR}(\boldsymbol{k}_*) \equiv \frac{\|\mathbb{E}[\textit{Signal}]\|^2}{\text{Tr}(\text{Cov}(\textit{Interference}))} = \frac{\mathcal{O}(T^{-2})}{\mathcal{O}(T^{-1})} = \mathcal{O}\left(\frac{1}{T}\right) \tag{18}$$

*Correspondingly, the amplitude Signal-to-Noise Ratio (*ASNR*) decays as:*

$$\mathrm{ASNR}(\boldsymbol{k}_*) \equiv \frac{|\mu_T|}{\sqrt{\mathrm{Var}(\boldsymbol{\xi}_T)}} \leq \frac{\sqrt{\boldsymbol{\phi}(\boldsymbol{k}_*)^\top \boldsymbol{\Sigma}_K^{-1} \boldsymbol{\phi}(\boldsymbol{k}_*)}}{\sigma_v \sqrt{T}} = \mathcal{O}\left(\frac{1}{\sqrt{T}}\right) \tag{19}$$

Corollary 1 (Asymptotic Associative Retrieval Failure)**.** *Consider a binary hypothesis test or codebook discrimination task where the model must decode target value $\boldsymbol{v}_* \in \{-1, +1\}$ from $\hat{\boldsymbol{y}}_*$ in the presence of Gaussian or sub-Gaussian interference. For any minimum detection margin $\gamma > 0$, the probability of classification error satisfies:*

$$P_{error} \geq 1 - \Phi\left(\mathrm{ASNR}(\boldsymbol{k}_*)\right) = 1 - \Phi\left(\frac{C}{\sqrt{T}}\right) \tag{20}$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. As context length $T \to \infty$:

$$\lim_{T\to\infty} P_{error} = 1 - \Phi(0) = \frac{1}{2} \quad (\textit{random guessing for binary bits}), \tag{21}$$

and for an M-ary codebook of vocabulary tokens ($M \gg 2$):

$$\lim_{T\to\infty} P_{retrieval\ error} = 1.0 \tag{22}$$

This proves why the Spectral Dilution Horizon cannot be solved by increasing numerical precision (e.g., from FP16 to FP32 or FP64): the degradation is an asymptotic geometric consequence of compressing an infinite sequence into a finite-rank subspace. Discrete token preservation via contiguous salience diffusion (HALO) is mathematically required for exact long-range recall.